

PROJECT SYNOPSIS

AI Agent-Based Indoor Virtual Campus Tour & Query Assistant

A multi-agent, RAG-grounded conversational assistant integrated with an immersive 360° virtual tour and interactive campus map, for Global Academy of Technology (GAT),
Bengaluru

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"Growing Ahead Of Time"

1. Abstract

The **GAT Virtual Campus** project is a web-based platform that lets prospective students, parents, and visitors explore Global Academy of Technology remotely and get accurate, grounded answers to their questions about the institution — before ever setting foot on campus. It combines four subsystems built on a shared codebase: an immersive 360° virtual tour of the Main Building navigable scene-by-scene or via an automated guided walk; an interactive satellite campus map; a Retrieval-Augmented Generation (RAG) AI assistant that answers admissions, academics, facilities, and fee questions strictly from GAT's own official documents and never invents information it cannot verify; and a multi-language interface (English, Kannada, Hindi) covering both the static UI and the AI assistant's spoken and generated responses.

The system is deliberately architected to keep two concerns separate that are often conflated in similar projects: a small trained intent-classification model is used only to help route a query to the right specialist, while a locally-hosted large language model (Llama 3.2 via Ollama) is solely responsible for composing the final natural-language answer, always grounded in retrieved, citable source text. No answer is ever generated purely from the model's own training data.

2. Introduction

Choosing an engineering college is a decision most students and parents make with very limited first-hand exposure to the campus itself. Physical visits are constrained by distance, time, and admission-season logistics, and a typical institutional website answers only the questions its authors anticipated, in a fixed language, with no way to ask a follow-up. Meanwhile, campus wayfinding for a first-time visitor — locating a specific department, laboratory, or office once they do arrive — remains a real, everyday friction point that generic maps do not solve for indoor, multi-floor buildings.

This project addresses both problems together: a virtual, navigable representation of the physical campus, and a conversational assistant that can answer specific, factual questions about admissions, fees, academics, and facilities — grounded in GAT's actual published information rather than a generic or hallucinated response.

3. Problem Statement

- Prospective students and parents cannot physically visit the campus before applying, and static websites/brochures cannot answer specific, personalized questions.
- Generic AI chatbots, if used naively, are prone to **hallucinating** facts — a serious risk for admission-critical information such as fee amounts, eligibility criteria, or deadlines.
- Indoor navigation (finding a department, lab, or room inside a multi-floor building) is not solved by standard outdoor mapping tools.
- A large fraction of GAT's prospective audience is more comfortable in Kannada or Hindi than English, and existing digital campus tools are English-only.

4. Objectives

1. Build an immersive, browser-based 360° virtual tour of the campus with both free (manual) exploration and an automated guided-walk mode.
2. Build an interactive campus map for outdoor orientation and building-level context.
3. Build a conversational AI assistant that answers questions **only from verified, official GAT sources** (retrieved documents, a structured fee database, and a small set of human-curated answers), refusing to answer rather than guessing when no reliable source exists.
4. Support voice input (speech-to-text) and spoken answers (text-to-speech) for accessibility and hands-free use.
5. Support English, Kannada, and Hindi across the interface and the AI assistant's generated answers.
6. Route each query to the right specialist (admissions, academics, facilities, navigation, general) using a lightweight intent classifier layered over deterministic keyword rules.

5. Scope of the Project

In scope: the Main Building's real, photographed interior (360° tour); satellite/outdoor campus map; an AI assistant grounded in scraped official GAT web pages/PDFs, a structured fee-and-admissions database, and curated FAQ answers; multi-language UI and AI responses; browser-native voice input/output; indoor pathfinding between surveyed campus nodes.

Out of scope (by design, for this phase): panoramic coverage of buildings beyond the Main Building; a mobile native app (the site is responsive web only); payment/transaction

processing; a login/authentication system (the assistant is anonymous, session-scoped only); user-generated content.

6. Technology Stack

Layer	Technology
Frontend framework	Next.js 15 (App Router), React 19, TypeScript (strict mode)
Styling / UI	Tailwind CSS, Framer Motion, lucide-react icons
State management	Zustand (client state), TanStack Query (server state)
360° panorama viewer	Pannellum
Interactive map	Google Maps JavaScript SDK (satellite/hybrid view)
Voice	Web Speech API — SpeechRecognition (STT) & SpeechSynthesis (TTS), browser-native
Backend framework	FastAPI (Python), Pydantic
Relational database	PostgreSQL, via SQLAlchemy ORM + Alembic migrations
Vector database	ChromaDB (persistent, local)
Embedding model	all-MiniLM-L6-v2 (Sentence-Transformers)
Lexical retrieval	BM25 (rank_bm25), fused with dense vector search
RAG orchestration	LangChain
LLM (answer generation)	Llama 3.2, served locally via Ollama — no external API, no per-token cost
Intent classifier	PyTorch LSTM (embedding → LSTM → dense → softmax), 12 intent classes
Multi-agent orchestration	Hand-rolled Python Supervisor + specialist agents (admissions, academics, facilities, navigation, general)
Pathfinding	A* search over a surveyed node/edge campus graph
Containerization	Docker Compose (frontend, backend, database, Ollama services)

7. System Architecture

The system follows a layered client–server architecture. The Next.js frontend never talks to the database or the language model directly; every dynamic request goes through the FastAPI backend's versioned REST API (`/api/v1/...`). Two independent data stores back the system: PostgreSQL holds structured, exact facts (campus graph, fee records, chat history), while ChromaDB holds embedded text chunks for semantic retrieval — each used for what it is good at, rather than forcing one store to do both jobs.

Chat request flow: User query → Supervisor (rule + LSTM intent routing) → specialist agent → Hybrid Retrieval (BM25 + dense vector fusion) → Reranking → Confidence scoring → *[if confident]* grounded LLM generation, or *[if not]* curated-answer semantic match, or a transparent "information unavailable" refusal → response with cited sources returned to the frontend.

8. Module Description

8.1 Virtual Tour Module

A scene-graph of real, photographed 360° panoramas across the Main Building's floors, connected as a doubly-linked structure per floor with stitched cross-floor transitions — enabling both free manual exploration (Next/Previous, hotspot clicks, minimap) and an automated Guided Tour mode that walks the visitor through the building phase-by-phase with orientation cues.

8.2 Interactive Campus Map Module

A satellite/hybrid map view of the campus footprint for outdoor orientation, complementing the indoor 360° tour.

8.3 AI Assistant (RAG) Module

The core of the project. User questions are answered strictly from retrieved context — never from the language model's own unguided knowledge. Retrieval combines lexical (BM25) and semantic (dense vector) search, fused and reranked, with a confidence score gating whether generation is even attempted. A structured **Fee & Admissions Knowledge Base** (PostgreSQL) holds officially-sourced, per-department tuition figures with explicit "not publicly disclosed" markers where no verified amount exists, so the assistant never invents a fee. A small **curated-answer** layer (semantically matched, not exact-string matched) serves as a second fallback for well-known questions the document corpus does not cover well, before the system ever falls back to an honest refusal.

8.4 Multi-Agent Routing Module

A Supervisor classifies each query (via deterministic keyword rules, backed by a trained LSTM classifier as a fallback signal) and routes it to the appropriate specialist agent — admissions, academics, facilities, navigation, or general — every one of which is grounded in the same retrieval pipeline, with routing decisions logged for traceability.

8.5 Navigation Module

Indoor pathfinding using the A* search algorithm over a surveyed graph of campus nodes and edges, returning turn-by-turn directions and estimated walking time between any two known locations.

8.6 Voice Interaction Module

Browser-native speech-to-text captures a complete spoken question (continuous recognition with silence detection, avoiding premature cut-off), and browser-native text-to-speech reads the assistant's answer aloud, with automatic voice selection and text cleaning (markdown/citation artifacts stripped from the spoken form only).

8.7 Multi-Language Module

A site-wide language selector (English default, Kannada, Hindi) drives static UI text, voice recognition/synthesis language, and the AI assistant's generation language — while deliberately keeping proper nouns (GAT, department names, VTU) and factual figures untranslated for accuracy.

9. Database Design (Summary)

PostgreSQL holds the campus graph (campuses → buildings → floors → rooms/nodes, with edges for pathfinding), panorama metadata, cross-floor hotspots, chat sessions/messages for conversational continuity, structured fee-and-admission records, and curated FAQ answers. ChromaDB separately stores embedded text chunks from official GAT PDFs and web pages for semantic search — the two stores are used together, not duplicated: structured facts (e.g. an exact fee figure) live in PostgreSQL as the source of truth, and the same fact is also rendered into a retrievable text chunk for natural-language question answering.

180

Campus graph nodes

325

Campus graph edges

161

Panorama scenes

1500+

Embedded KB text chunks

12

Intent classes

3

Interface languages

10. Methodology

The project was built incrementally, phase by phase, each phase independently runnable and demonstrable before the next began: frontend shell → knowledge-base construction and RAG pipeline → multi-agent routing → voice interaction → 360° tour and indoor navigation → multi-language support → evaluation. The knowledge base itself was built from real, scraped official GAT web pages and PDF documents (not a generic or unrelated dataset), chunked, embedded, and indexed — with a documented source manifest tracing every answer back to a specific official document or page.

11. Current Status / Results

- The RAG pipeline correctly answers real test queries (e.g. department fee amounts, admission process, required documents) with cited sources and a measured confidence level, and correctly **refuses** rather than fabricates an answer when asked about a fee for a department that does not exist.
- The trained LSTM intent classifier was evaluated on its held-out validation split with real, reproducible metrics (not simulated), including a full confusion matrix.
- End-to-end voice queries (speak → transcribe → retrieve → generate → speak the answer) function correctly, including in Kannada and Hindi where the browser/OS provides a corresponding voice.
- The guided and manual virtual tour, cross-floor navigation, and the interactive map all function correctly across the supported languages without disrupting existing tour mechanics.

12. Advantages

- **Grounded, not hallucinated:** every AI answer traces back to an official source, a structured database record, or a human-curated answer — never the model's own unverified training data.
- **No external API dependency** for the LLM — Ollama runs locally, with no per-query cost and no dependency on an external provider's uptime.

- **Accessible** across three languages and via voice, broadening who can realistically use the assistant.
- **Extensible knowledge base** — new official documents or curated answers can be added without retraining any model or redesigning the retrieval pipeline.

13. Limitations

- 360° photographic coverage is currently limited to the Main Building; other campus buildings are not yet photographed.
- Answer generation runs on a local, CPU-served language model, which is slower than a cloud API under load.
- Some fee categories (e.g. government/KCET-quota tuition, hostel fees) are not publicly published by any official GAT source, so the assistant correctly reports them as unavailable rather than estimating them.
- The intent classifier is trained on a small, synthetic seed dataset rather than large-scale real user query logs.

14. Future Scope

- Extend 360° photographic coverage to additional campus buildings.
- Expand the curated-answer and fee knowledge base as more official documents are published each academic year.
- Add streaming (token-by-token) responses for a faster-feeling chat experience.
- Formal evaluation of faithfulness/relevancy using an LLM-judge pipeline across a fixed benchmark question set.
- Outdoor GPS-guided navigation connecting the map view to the indoor tour.

15. Conclusion

The GAT Virtual Campus project demonstrates that a genuinely useful, trustworthy AI assistant for an institutional website is achievable without relying on an external, paid API and without accepting the hallucination risk that comes with an ungrounded chatbot — by combining structured data, retrieval-augmented generation, and honest refusal as first-class design decisions, alongside an immersive virtual tour and multi-language, voice-accessible

interface that together give prospective students and parents a realistic sense of the campus and reliable answers to the questions that matter most to them.